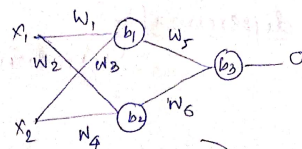


Activation Function

24/08/21

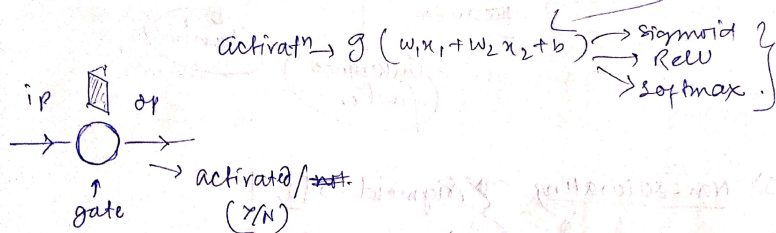
What are Activation Functions?

In artificial NN, each neuron form a weighted sum of its inputs and passes the resulting scalar value through a function referred to as an activation function. If a neuron has n inputs then the output or activation of a neuron is —.

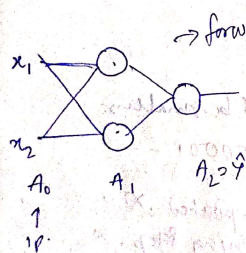


$$a = g(w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n + b)$$

This fn 'g' is referred to as the activation function.



Why activation functions are needed?



forward propagation

$$Z_1 = w_1A_0 + b_1$$

$$A_1 = g(Z_1) = Z_1$$

linear fn.

$$A_2 = g(w_2A_1 + b_2) = w_2A_1 + b_2$$

$$= g(w_1A_0 + b_1)$$

$$= w_2(w_1A_0 + b_1) + b_2$$

$$= w_2w_1A_0 + w_2b_1 + b_2$$

$$= w'A_0 + b'$$

$$A_2 = \hat{y} = w'A_0 + b'$$

$A_2 \propto A_0 \rightarrow$ linear Relationship

Nonlinear Activation fns helps to capture Non-linearity of the data. In ANN.

Without Activation function Example

Ideal Activation fn

1) Non-linear

$y = f(x) \rightarrow$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

2) differentiable

$\rightarrow$ gradient descent

$\rightarrow$ derivatives $\rightarrow$ relu(0)

3) Computationally inexpensive

$\rightarrow$ derivative $\rightarrow$ simple/easy fast

4) Zero centered

$\rightarrow$ zero centered

$\rightarrow$ Normalized

O/p $\rightarrow$ same dirn of update of gradient descent?

mean 20

converge faster

$\rightarrow$ Normalized data later layers input

tanh

5) Non-saturating

Derivative of o/p $\rightarrow$ caused Vanishing Gradient Problem?

Sigmoid $\rightarrow [0, 1]$

tanh $\rightarrow (-1, 1)$

saturating activation functions

ReLU $\Rightarrow f(x) = \max(0, x) \rightarrow$ non saturating

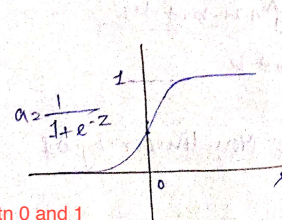
Problem

backpropagation $\rightarrow$ value will be smaller

$0.01 \rightarrow 0.0001 \rightarrow 0.000001 \dots$

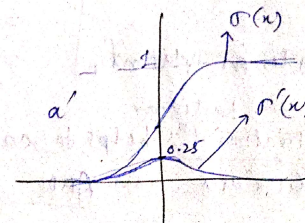
weights not updated. $\times$ training stop. $\times$

Sigmoid Activation function



O/p betn 0 and 1

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



$$\sigma'(x) = \sigma(x) \cdot (1 - \sigma(x))$$

Maxm derivative at $x=0$ $\sigma'(x)=0.25$

Advantages

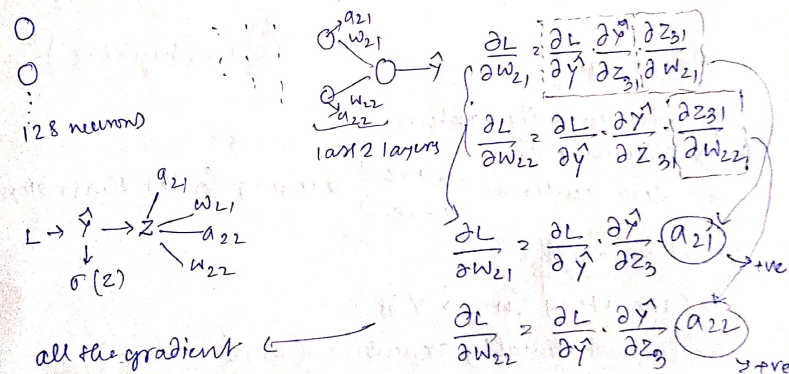
- 1) $[0, 1] \rightarrow$ probability $\rightarrow$ output layer $\rightarrow$ Binary classification.
- 2) Non-linearity $\rightarrow$ Non-linear data $\rightarrow$ good option.
- 3) Differentiable $\rightarrow$ Backpropagation $\rightarrow$ $\left[\frac{\partial L}{\partial w} \right]$.

Disadvantages

- 1) saturating function $\rightarrow [-\infty, \infty] \rightarrow [0, 1]$
 $\rightarrow$ vanishing Gradient problem
 Backprop $\rightarrow$ update xx .

2) Non Zero Centered

- Normalized $\rightarrow$ +ve $\rightarrow$ Training slow
- $\downarrow$ mean $\neq 0$ $\rightarrow$ -ve $\rightarrow$ convergence problem.



all the gradient can be all +ve or all -ve in the same time.

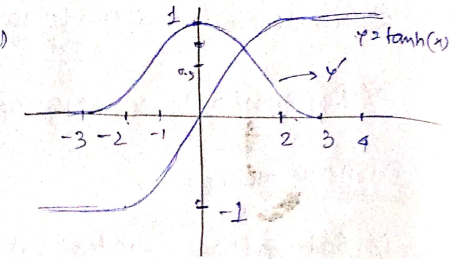
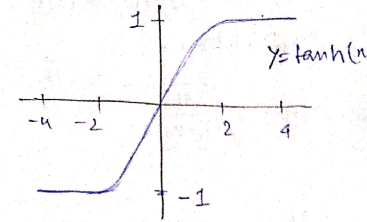
means the weight update/charge done through all +ve or all -ve. Means all +ve updates or all -ve updates.

$$\begin{cases} 0+ve \\ 0+ve \\ 0+ve \end{cases} \rightarrow \text{Restriction} \begin{cases} 0+ve & 0-ve \\ 0+ve & 0-ve \\ 0+ve & 0-ve \end{cases}$$

- 2) $\sigma(x) = \frac{1}{1+e^{-x}}$ $\rightarrow$ computationally expensive because of the exponent.

2) Tanh-Activation function

Hyperbolic tangent



$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\begin{aligned} f'(x) &= \frac{(e^x - e^{-x})(e^x + e^{-x}) - (e^x + e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2} \\ &= \frac{(e^{2x} - e^{-2x}) - (e^{2x} - e^{-2x})}{(e^x + e^{-x})^2} \\ &= 1 - \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)^2 \\ &= 1 - (\tanh(x))^2 \end{aligned}$$

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

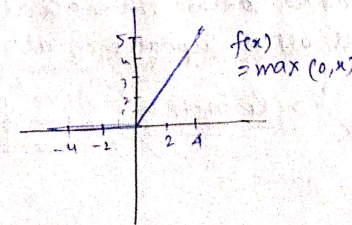
Advantages

- 1) Non-linearity captures
- 2) Differentiable
- 3) Zero-centered $\rightarrow$ +ve $\rightarrow$ -ve $\rightarrow$ training faster than Sigmoid.

Disadvantages

- 1) Saturating fn $\rightarrow$ VGP.
- 2) Computationally expensive (expo $\rightarrow$ slow).

3) ReLU-Activation Function



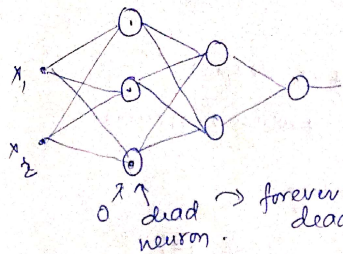
Advantages

- 1) Non-linear
- 2) Not saturated in +ve region
- 3) Computationally inexpensive
- 4) Convergence $\rightarrow$ faster than sigmoid & tanh

Disadvantages

- 1) not completely differentiable $|x=0|$ $\rightarrow$ assumption $x < 0 \rightarrow 0$ $x > 0 \rightarrow 1$
- 2) Not zero centered $\rightarrow$ to solve $\rightarrow$ Batch Normalization

Dying ReLU Problem



Relu $\rightarrow$ Rectified Linear Unit.

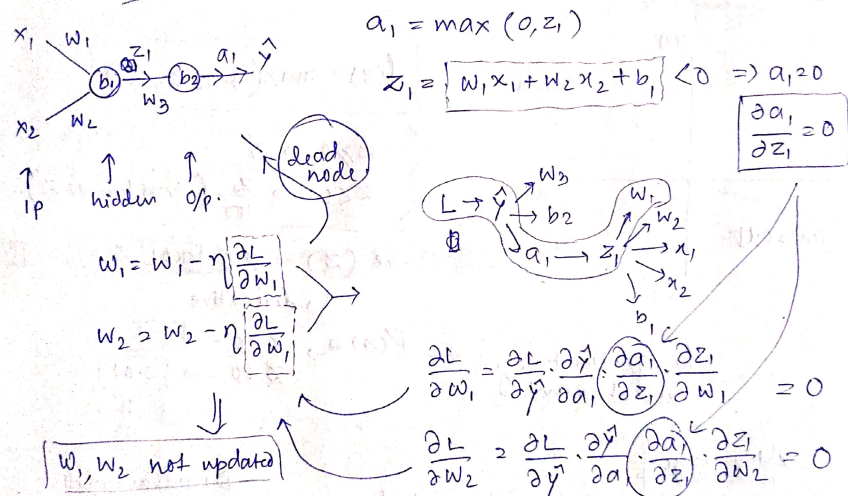
Best choice hidden layers

Don't can $\rightarrow$ 100% neurons dead
 $\rightarrow$ No network exist

any input $\rightarrow$ always 0/p $\rightarrow$ 0 (Zero)
 cannot capture patterns of the data
 gives low level representation

negative output in relu is problem because it will make it zero, making dead neurons

Why?



$z_1 = w_1 x_1 + w_2 x_2 + b_1 < 0 \rightarrow$ when possible?

① High learning rate $\rightarrow$ $w_1 = w_1 - \eta \frac{\partial L}{\partial w_1}$ $\rightarrow$ overall very high

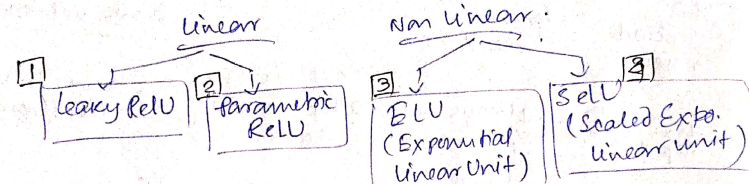
So in the next cycle z_1 can be < 0
 $z_1 = w_1 x_1 + w_2 x_2 + b_1$
 $\rightarrow$ Dying ReLU Problem.

② high -ve bias $\rightarrow$ $w_1 x_1 + w_2 x_2 + b_1$ highly -ve. can be a reason

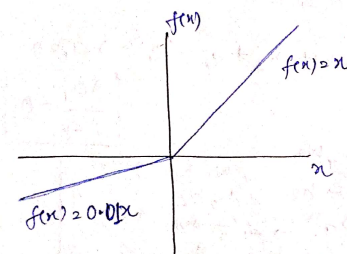
another reason $\left\{ \begin{array}{l} b_1 = b_1 - \eta \frac{\partial L}{\partial b_1} \\ \rightarrow \text{if this is highly -ve. or applying high } \eta \text{ can lead to this.} \end{array} \right.$
 $\rightarrow$ the total expression can be -ve.

Solutions

- ① Set low learning Rate.
- ② bias $\rightarrow$ +ve value $\rightarrow$ [0.01]
- ③ Don't use Relu $\rightarrow$ variants. (+ve keep) problems $\rightarrow$ remove.



1 Leaky ReLU



$$f(z) = \max(0.01z, z)$$

$$\begin{cases} z \geq 0 \rightarrow z \\ z < 0 \rightarrow \frac{z}{100} \text{ (fraction of } z) \end{cases}$$

-ve (z) $\rightarrow \neq 0$ [0.01] z
 $\rightarrow$ derivative

the gradient cannot become 0. $\leftarrow f'(z) \rightarrow \begin{cases} z \geq 0 \rightarrow 1 \\ z < 0 \rightarrow 0.01 \end{cases}$

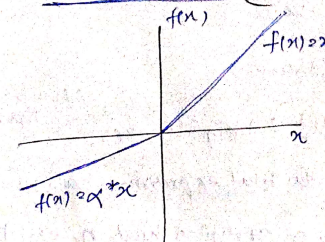
Advantages

- ① Non-saturated $\rightarrow$ unbounded
- ② Easily computed
- ③ No dying ReLU Problem.
- ④ Close to zero-centered. (-ve/+ve)

Disadvantages

why only 0.01?
 ??

2 Parametric ReLU (PReLU)

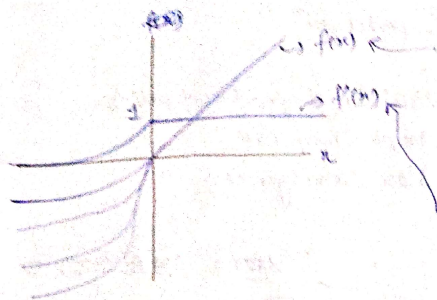


$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ ax & \text{otherwise} \end{cases}$$

$a \rightarrow$ trainable parameter
 flexibility.

(all other $\rightarrow$ exactly same)

5 ELU - Exponential Linear Unit



$$ELU(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(e^x - 1) & \text{if } x \leq 0 \end{cases}$$

always continuous & always differentiable

$$ELU'(x) = \begin{cases} 1 & \text{if } x > 0 \\ ELU(x) + \alpha & \text{if } x \leq 0 \end{cases}$$

constant.

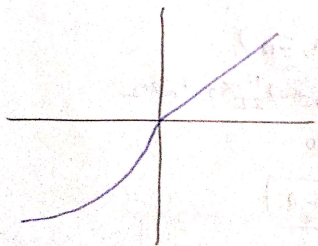
Advantages

- 1 close to zero centered
↳ convergence faster
- 2 better generalized
- 3 dying relu x
- 4 Always continuous & differentiable.

Disadvantages

computationally expensive.
(ex).

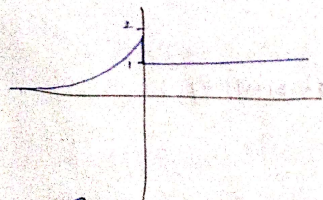
7 SELU - Scaled Exponential Linear Unit



$$SELU(x) = \lambda \begin{cases} x & \text{if } x > 0 \\ \alpha e^x - \alpha & \text{if } x \leq 0 \end{cases}$$

$$\alpha \approx 1.6733, \lambda \approx 1.0507$$

fixed parameters



$$SELU'(x) = \lambda \begin{cases} 1 & \text{if } x > 0 \\ \alpha e^x & \text{if } x \leq 0 \end{cases}$$

Extra Advantage

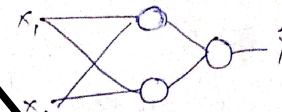
- 1 self normalizing → activation
↓
normalized
↓
μ=0, std=1
converge faster

Disadvantage

- 1 Research ongoing.

25/07/23 Weight Initialization Techniques

Why weight initialization is important?



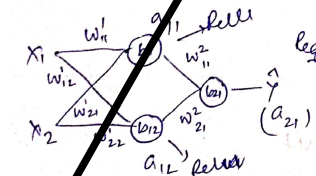
- vanishing gradient problem
- exploding GP.
- slow convergence.

Wrong wt. init.

What to do?

Wrong techniques:

Case 1: Zero Initialization



Regression Example:

$$a_{11} = \max(0, z_{11})$$

$$z_{11} = w'_{11}x_1 + w'_{21}x_2 + b_{11}$$

$$a_{12} = \max(0, z_{12})$$

$$z_{12} = w'_{12}x_1 + w'_{22}x_2 + b_{12}$$

$$a_{11} = 0, a_{12} = 0$$

$$a_{11} = a_{12} = 0$$

$$w = 0, b = 0$$

$$w'_{11} = w_{11} - \eta \frac{\partial L}{\partial w_{11}} = 0$$

no update. No training will take place.

$$a_{11} = \frac{e^{z_{11}} - e^{-z_{11}}}{e^{z_{11}} + e^{-z_{11}}} = 0 \rightarrow \text{no update} \rightarrow \text{No training.}$$

Sigmoid.

$$a_{11} = \sigma(z_{11}) = 0.5$$

$$z_{11} = 0$$

$$a_{12} = \sigma(z_{12}) = 0.5$$

$$z_{12} = 0$$

$$\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-0}} = \frac{1}{2} = 0.5$$

$$w = w - \eta \frac{\partial L}{\partial w}$$